Classification Models

Applied Machine Learning — Session 2, Chapter 2

Classification: What We're Doing

Predicting a discrete category.

- Binary: Spam / Not Spam; Disease / Healthy
- Multi-class: Iris species; Digit recognition (0–9)

What the model outputs:

- Hard prediction: `model.predict(X)` → class label
- Probability: `model.predict_proba(X)` → [P(class 0), P(class 1), …]

Logistic Regression

A classification model, not regression!

Models the probability of class membership:

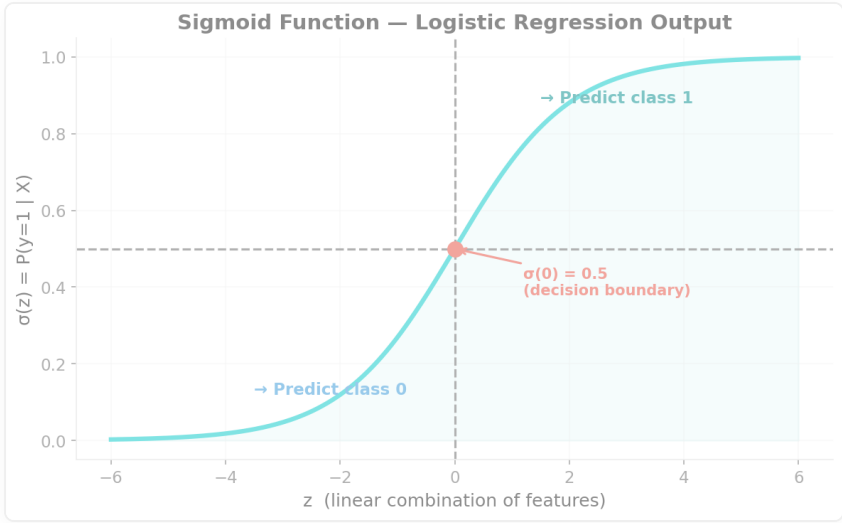
$$P(y=1 \mid X) = \sigma(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)$$

where σ is the sigmoid function: σ(z) = 1 / (1 + e^-z)

```python
from sklearn.linear_model import LogisticRegression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
proba = model.predict_proba(X_test)  # ← probabilities
```

Linear decision boundary — fast, interpretable, great baseline.

The Sigmoid Function



- σ(0) = 0.5 → decision boundary
- σ(very large) → 1
- σ(very negative) → 0

Decision: If P(y=1) > 0.5 → predict class 1

K-Nearest Neighbors (KNN)

"Tell me who your neighbors are, and I'll tell you what you are."

Algorithm:

1. Find the k closest training samples (by distance)
2. Predict the majority class among those k neighbors

```
from sklearn.neighbors import KNeighborsClassifier
knn = KNeighborsClassifier(n_neighbors=5)
```

⚠️ **Requires feature scaling** — distance is scale-sensitive!

⚠️ Slow at prediction time for large datasets

Choosing k in KNN

```
k = 1 → very complex, overfitting
k = large → very smooth, underfitting
k = √n → common rule of thumb
```

Always cross-validate to find the best k!

```python
from sklearn.model_selection import cross_val_score
for k in [1, 3, 5, 7, 10]:
    score = cross_val_score(KNeighborsClassifier(k), X, y, cv=5).mean()
    print(f'k={k}: {score:.3f}')
```

Decision Tree Classifier

Learns a series of yes/no questions.

```
Is petal_length < 2.5?
   YES → Setosa ✓
   NO  → Is petal_width < 1.8?
              YES → Versicolor ✓
              NO  → Virginica ✓
```

```python
from sklearn.tree import DecisionTreeClassifier, plot_tree
dt = DecisionTreeClassifier(max_depth=3)
dt.fit(X_train, y_train)
plot_tree(dt, feature_names=feature_names, filled=True)
```

Perfectly interpretable — you can explain every decision.

Random Forest Classifier

Many trees → better, more robust predictions.

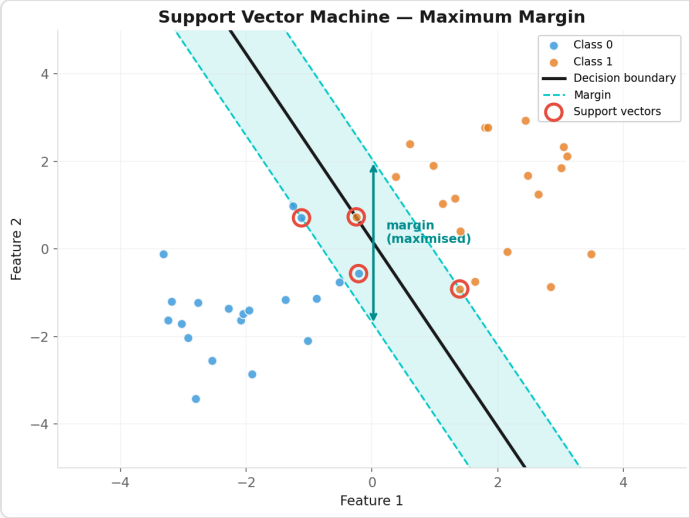
```
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

# Which features matter most?
importances = pd.Series(rf.feature_importances_, index=feature_names)
importances.sort_values().plot(kind='barh')
```

Ensemble power:

- Reduces variance (averaging reduces noise)
- More robust to outliers
- Less prone to overfitting than single DT

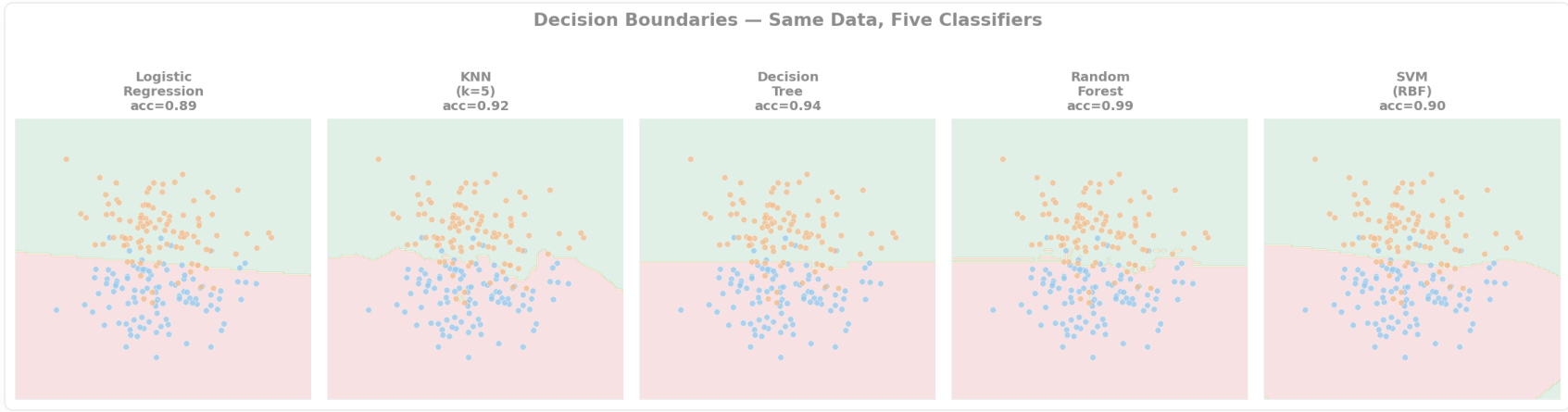
Support Vector Machines (SVM)

Find the hyperplane with maximum margin.




```
from sklearn.svm import SVC
svm = SVC(kernel='rbf', C=1.0, probability=True)
```

Decision Boundaries: What Each Model Learns



→ See examples notebook for visual comparison!

Now: Exercises!

→ Open `03-exercises/ch05_classification_exercises.ipynb`

Task: Classify wine types using multiple classifiers.

Apply multiple models, compare performance.

~12 minutes

Key Takeaways

- Logistic Regression: linear, interpretable, probabilistic
- KNN: simple, local, needs scaling
- Decision Tree: rule-based, interpretable, overfits easily
- Random Forest: robust ensemble, feature importances
- SVM: maximum margin, powerful with kernels

Next: Chapter 6

Metrics & Evaluation

"Which model is actually better? It depends on how you measure it."